

Where the Number Helix Lives, and How to Find It: Cross-Architecture and Cross-Script Geometry of Integer Representations

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Abstract

Kantamneni and Tegmark [1] showed that for an integer $a \in [0, 99]$, the residual stream of three mid-sized LLMs (GPT-J-6B, Pythia-6.9B, Llama-3.1-8B) encodes a as a generalised helix—a linear “number-line” axis plus four orthogonal modular circles at periods $T \in \{2, 5, 10, 100\}$, and that this helix is causally used to perform addition. Their basis itself is the LLM-scale analogue of Nanda et al. [2], who reverse-engineered the same trig solution in a small transformer trained from scratch on modular addition—the canonical grokking task of Power et al. [3]. We extend the LLM result along three orthogonal axes. **Architecture:** we add two new model families (Gemma 4 at E4B and 31B; OLMo-3 at 32B) and re-measure Pythia and Llama under a wider protocol, recovering the helix on all five models but at radically different depths—Pythia/Llama mid-stack, Gemma-4-E4B at $L=5$ of 34, Gemma-4-31B as a sharp spike at $L=29$ of 61, OLMo-3-32B at $L=23$ of 64 with a separate early helix/PCA peak at $L=4$. **Script:** across eight numeral systems we find a three-way split—positional base-10 scripts the model has been trained on (Latin, Devanagari, Arabic-Indic, Persian, positional Chinese) reproduce the helix on multilingual models; additive systems (Roman, Greek alphabetic) collapse it in two qualitatively distinct ways distinguishable only by inspecting where PC1 and helix R^2 peak in layer-space; and Babylonian cuneiform, a positional base-60 system, encodes its native period 60 on *all five models*, provided we widen the input window past one base-60 wrap and add $T = 60$ to the trig basis (adding the single missing period recovers 7–13 percentage points of variance uniformly across the five models). 2-D PCA scatters of the residual stream further reveal that on both Gemma and OLMo the first principal component is not magnitude but *tokenisation* (PC1 captures up to 96.6% of variance but separates points by digit count, not value), a pattern absent only on Pythia. We extract two methodological lessons: (M1) any basis-dependent metric must be cross-checked against a basis-free FFT and against a 2-D PCA scatter; (M2) the input range must contain several wraps of every candidate period. A third diagnostic shifts the headline reading substantially: comparing helix R^2 at the embedding layer ($L = 0$, before any transformer block) to its peak across all layers reveals that most non-Latin and all Babylonian cells in our matrix already carry their reported helix at $L = 0$ —the transformer adds little. Only Pythia/Latin, Llama/Latin (the original paper’s cells) and the Gemma cells on Devanagari, Persian, and Roman show transformer blocks doing the majority of the representational work. The cross-script and Babylonian generalisations are therefore most honestly read as tokenisation + mean-pooling artifacts rather than learned arithmetic representations.

1 Introduction

Mechanistic interpretability has produced a growing catalogue of “cognitive” structures embedded in transformer residual streams—induction heads, indirect-object identification circuits, and so on. Arithmetic has been a particularly load-bearing case: Power et al. [3] observed

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that a small transformer trained on modular addition continues training long past memorisation before suddenly generalising (“grokking”), and Nanda et al. [2] reverse-engineered the resulting solution as a Fourier/trig basis—integers laid out on circles, addition implemented via $\cos(a + b) = \cos a \cos b - \sin a \sin b$. The number-helix of Kantamneni and Tegmark [1] carries the same basis into *pretrained* LLMs on the naturals: integers are not stored as opaque embeddings but laid out as a 3-D spiral—a linear number-line axis plus four orthogonal circles—and addition is performed by rotating the spiral’s circular components. The claim is striking; it is also based on three models in the same broad decoder-only lineage (GPT-J-6B, Pythia-6.9B, Llama-3.1-8B) and on a single numeral system (Latin digits).

This paper asks: *how general is the helix?* We ask the question along two independent axes:

1. **Architecture:** does the helix extend beyond the decoder-only lineage Kantamneni and Tegmark studied? We add Gemma 4 (E4B and 31B) and OLMo-3-32B (AllenAI’s fully-open 32B)—each with a different tokeniser, training mix, and depth—and re-measure Pythia and Llama under a wider protocol. If the helix is there, at what depth?
2. **Script:** when the model reads “۲۳”, “۳۳”, “<<III”, does it represent these as numbers in the same helical way, or as ordinary multi-byte words?

Both axes turn out to matter, and both surface methodological subtleties that are easy to miss if one runs only the paper’s exact measurement recipe. Section 2 reviews the helix construction and our two basis-free diagnostics. Section 3 reports the layer-depth findings across five models. Section 4 compares eight numeral systems, and analyses the two qualitatively distinct ways the helix collapses on non-positional scripts. Section 5 contains the main scientific surprise: Babylonian cuneiform is represented at its native base 60 on every model we tested, once the measurement window is widened past a single wrap and the basis is extended with $T = 60$. Section 6 introduces our third diagnostic—the ratio of helix R^2 at $L=0$ to its peak—which reveals that on most non-Latin cells, the helix is already there before the transformer runs, and substantially revises which of the cross-script results should be read as “the model represents this” versus “the tokeniser plus pooling produce a helix-fittable shape.” Section 7 distills the two generalisable methodological lessons.

2 Methods

2.1 Setup

For each integer $a \in [0, n_{\max})$ we feed the string “ $\lceil \text{numeral}(a) \rceil$ ” to the model and pool the residual stream at a chosen layer L across the sub-tokens of the numeral. With d the hidden width, the result is an activation matrix $H \in \mathbb{R}^{n_{\max} \times d}$. Latin digits 0–99 are single tokens on Pythia, so pooling is trivial there; in every other case (non-Latin scripts; Latin on Llama and Gemma; the $n_{\max}=600$ sweep) we average activations across the numeral’s sub-tokens to avoid a last-byte cycling artifact in which $\{23, 33, 43, \dots\}$ read identical activations because they end in the same glyph.

2.2 The helix basis

The Kantamneni–Tegmark hypothesis is that $h(a)$, for fixed L , factors as

$$h(a) \approx W^\top B(a), \quad B(a) = [a, \cos \frac{2\pi a}{T_1}, \sin \frac{2\pi a}{T_1}, \dots, \cos \frac{2\pi a}{T_K}, \sin \frac{2\pi a}{T_K}], \quad (1)$$

with $T = \{2, 5, 10, 100\}$ for Latin in $[0, 99]$, giving a $(1 + 2K) = 9$ -dimensional feature space. The linear coordinate is the “rise” of the helix; each $(\cos, \sin)$ pair places a on the unit circle at

angle $2\pi a/T$, so integers sharing the same $a \bmod T$ land on the same point. Addition becomes rotation of angles: the paper’s “Clock” algorithm.

We solve $H \approx BW$ by ordinary least squares and report the variance-weighted coefficient of determination, which we call *helix* R^2 .

2.3 Three diagnostics

Helix R^2 alone is fragile—it depends on the fixed basis T . We pair it with two basis-free diagnostics.

Fourier diagnostic. For each hidden dimension j , the column $H_{:,j}$ is a sequence indexed by a . We take its discrete Fourier transform, average the magnitudes across j , and inspect the spectrum. Periodic structure *at any period*, including ones not in T , surfaces as a peak. We label the top peaks via `scipy.signal.find_peaks` so that the FFT panel honestly reports whatever periods the model actually encodes, rather than only those the paper’s basis assumes.

PC1 diagnostic. The leading principal component of H is the direction of maximum variance. If the helix’s linear axis is present, plotting PC1 against a should yield a clean line; we report the linear-fit R^2 as *PC1* R^2 .

Eckart–Young upper bound. Helix R^2 uses a 9-D fit, but 9 is also the dimensionality of many other regression targets. The best possible 9-D fit is the 9-D PCA reconstruction; we call its variance-weighted R^2 the *K-D PCA bound*, and report the ratio $\rho = R_{\text{helix}}^2/R_{\text{PCA}}^2$ as *helix/PCA*. When $\rho \gtrsim 0.8$, the trig basis is *the* 9-D structure of the residual stream rather than merely a 9-D approximation of it.

2.4 Layer sweep

The paper reads at layer $L = N/2$ in an N -layer model. We capture activations at all $L = 0, 1, \dots, N$ in one forward pass per integer and compute (PC1 R^2 , helix R^2 , *K-D PCA bound*) at each. The peak helix- R^2 layer is then used for the standard regression fit. As Section 3 shows, “read the middle layer” is a Pythia/GPT-J convention, not a universal one.

2.5 2-D PCA scatter

At the helix-peak layer of each cell we scatter the first two principal components of H , colour-coded by a , with consecutive integers connected by a faint trajectory line. This is the cheapest basis-free diagnostic for *non-periodic* structure (Roman’s threshold-aligned clusters; Greek’s diffuse noise; Gemma’s tokenisation-dominated PC1; Section 4.5).

3 Architecture: the helix is consistent, the layer is not

We test five base (non-instruction-tuned) models: Pythia-6.9B, Llama-3.1-8B, Gemma-4-E4B (efficient ~ 4 B), Gemma-4-31B, and OLMo-3-32B (AllenAI’s fully-open 32 B). All five converge on the same geometric encoding on Latin digits 0–99 (Table 1), but the depth at which the helix peaks varies dramatically, and so does the shape of the depth profile (Figure 1).

Four depth regimes emerge:

- **Pythia and Llama:** helix forms in the back half of the stack and remains until at least $L = N - 1$. Either $L = N/2$ or $L = N - 1$ would have worked.
- **Gemma-4-E4B:** helix is strongest at $L = 0-5$ and decays in later layers. The paper’s middle-layer recipe would literally read at decayed structure.

Table 1: Latin helix metrics at the peak layer of each model. “helix/PCA” is the ratio of helix R^2 to the 9-D PCA upper bound (values $\gtrsim 0.8$ indicate the trig basis dominates the 9-D subspace).

Model	helix R^2	helix/PCA	peak layer	total layers
Pythia-6.9B	0.53	0.85	15	32
Llama-3.1-8B	0.63	0.85	14	32
Gemma-4-E4B	0.50	0.77	5	34
Gemma-4-31B	0.70	0.75	29	61
OLMo-3-32B	0.42	0.83	23	64

- **Gemma-4-31B**: helix R^2 is essentially flat at ~ 0.5 across the whole stack, with a single sharp peak of 0.70 at $L = 29$. The default $L = N/2 = 30$ misses the peak by exactly one layer.
- **OLMo-3-32B**: helix R^2 rises gradually to a mid-stack peak at $L = 23$ of 64 (helix $R^2 = 0.42$, helix/PCA = 0.83), with a second very-early peak in the helix/PCA ratio at $L = 4$ that resembles the Gemma-4-E4B early-helix profile. Two different geometric regimes coexist within the same stack.

We initially concluded “Gemma-4-31B does not form a helix” from a default-layer reading—a conclusion wrong by a single layer. The methodological consequence is unambiguous: *for any new model, a layer sweep is mandatory*. The helix is general; the recipe for finding it is not.

4 Script: positional structure is necessary, training exposure is what makes the helix appear

4.1 The numeral systems at a glance

For readers unfamiliar with the non-Latin systems we test:

- **Arabic-Indic** (٠١٢٣٤٥٦٧٨٩) and **Persian** (۰۱۲۳۴۵۶۷۸۹) are positional base-10 like Latin, differing only in glyph shape (Persian is a visual variant of Arabic-Indic). Numerals are written most-significant-first despite Arabic script flowing right-to-left.
- **Devanagari** (०१२३४५६७८९) is also positional base-10, used across Indian languages including Hindi, Sanskrit, and Marathi.
- **Positional Chinese** uses the digits 零一三三四五六七八九 in the same place-value layout as Latin (e.g. 二三 for 23). Classical Chinese also admits a non-positional form (二十三, “two-tens-three”); we do not use it here.
- **Greek alphabetic (Milesian)** is additive: each letter carries a fixed value ($\alpha = 1, \dots, \theta = 9, \iota = 10, \kappa = 20, \dots, \pi = 80, \rho = 90, \rho = 100, \dots$). 23 is $\kappa\gamma = 20 + 3$. There is no shared “ones digit” between 13, 23, and 33.
- **Roman** is additive with subtractive shortcuts at 4, 9, 40, 90, etc.: I= 1, V= 5, X= 10, L= 50, C= 100. 23 = XXIII. Glyphs shift at $4 \leftrightarrow 5$, $9 \leftrightarrow 10$, $49 \leftrightarrow 50$, $89 \leftrightarrow 90$ — the same boundaries that later appear as staircase discontinuities in the PC1 panel (Section 4.4).
- **Babylonian cuneiform** is positional at base 60, additive within each sexagesimal column. Two glyphs combine inside a column: $\uparrow$ (one) and $\leftarrow$ (ten). 23 is $\leftarrow\leftarrow\uparrow\uparrow$ — two tens plus three ones, all in one column. 60 opens a new column to the left (Section 5).

4.2 Setup

We rendered every integer in $[0, 99]$ in eight numeral systems (Table 2) and re-ran the analysis under mean-pooling.

Table 2: Numeral systems tested. The Babylonian column is the most interesting and is treated separately in Section 5.

Script	Example (23)	Family
Latin	23	positional base-10
Arabic-Indic	٢٣	positional base-10
Persian	۲۳	positional base-10
Devanagari	२३	positional base-10
Chinese (pos.)	二三	positional base-10
Greek alphabetic	κγ	additive (Milesian)
Roman	XXIII	additive
Babylonian	<<III	positional base-60, additive within column

4.3 Helix/PCA across scripts

Table 3: Helix/PCA ratio at each model’s peak layer, by script (mean pooling). Bold = highest score per model.

Script	Pythia-6.9B	Llama-3.1-8B	Gemma-4-E4B	Gemma-4-31B
Latin	0.85	0.85	0.77	0.75
Arabic-Indic	0.55	0.58	0.66	0.69
Persian	0.55	0.70	0.72	0.72
Devanagari	0.53	0.61	0.78	0.79
Chinese	0.60	0.58	0.68	0.66
Greek (additive)	0.54	0.57	0.57	0.57
Roman (additive)	0.54	0.55	0.62	0.64

Three observations stand out:

Non-positional scripts collapse uniformly. On every model, Greek and Roman collapse the helix/PCA ratio to ~ 0.55 – 0.65 . A trig basis with a tens-loop only buys representational efficiency if the numeral system has a tens column. Roman IV versus V is not the kind of transition a fixed-period circle can align to.

Latin is uniform. Four of the five models clear 0.75 on Latin (OLMo-3-32B sits lower at helix/PCA 0.83 with absolute helix $R^2 = 0.42$). The shape itself is not model-specific.

Non-Latin positional cells split by training data. Pythia on Devanagari (0.53) is statistically indistinguishable from Pythia on Greek (0.54): the helix is gone. Llama on Devanagari (0.61) is a little better. Gemma-4-31B on Devanagari hits 0.79, matching its own Latin score of 0.75. Pythia’s training mix is English-heavy; Llama saw more multilingual data; Gemma is the most multilingual of the four. *Positional structure is necessary for the helix, but training exposure to the specific script is what makes it appear.* The encoding is built per-script during pre-training; architecture alone is not enough.

4.4 Two qualitatively distinct collapses on non-positional scripts

The naive reading of Table 3 is that “Greek and Roman collapse the helix.” That is correct on the numerical metric but misleading: the underlying geometry is qualitatively different in the two cases. Reading PC1 R^2 alongside helix R^2 , *and the layer at which each peaks*, reveals the difference (Table 4).

Table 4: Layer of peak PC1 and helix R^2 on Latin, Roman, Greek. Latin: both peak at the same layer (a single helix). Roman: peaks are 9–20 layers apart (decomposed parallel modules). Greek: helix R^2 peaks at $L = 0$ (tokeniser artifact at the embedding).

Model	Script	PC1 R^2 peak	helix R^2 peak	layer gap
Pythia-6.9B	Latin	0.96 @ L5	0.53 @ L32	—
Pythia-6.9B	Roman	0.66 @ L9	0.36 @ L18	9
Pythia-6.9B	Greek	0.10 @ L9	0.28 @ L0	(artifact)
Llama-3.1-8B	Roman	0.51 @ L16	0.36 @ L17	1
Llama-3.1-8B	Greek	0.17 @ L5	0.27 @ L0	(artifact)
Gemma-4-E4B	Roman	0.54 @ L4	0.61 @ L18	14
Gemma-4-31B	Roman	0.49 @ L11	0.64 @ L31	20

Roman: a decomposed encoding, not a fused helix. On Latin, PC1 and helix R^2 peak at the same mid-stack layer; magnitude and periodicity co-locate, and the trig basis literally fits a helix (line $\times$ circles). On Roman, PC1 and helix R^2 peak at *different* layers—9 to 20 apart. Magnitude is present (PC1 R^2 climbs to 0.49–0.66, so the model does coarsely order Roman numerals), and so is periodic structure (Figure 2 shows clean FFT peaks at $T = 10$ for the X-cycle and $T = 5$ for the V-cycle), but they live in different parts of the residual stream. Geometrically, Roman is encoded as a *number-line module* at one depth plus a *letter-cycling module* at another, not as a fused helix. The PC1-vs- a panel of Figure 2 is also informative: it is not a straight line but a staircase with discontinuities at the I/V, IX/X, XL/L, and XC/C boundaries—the underlying signal is piecewise linear, not linear, which is why the naive linear-fit $R^2 = 0.23$ understates the magnitude information that is present.

Greek: no quantitative representation at all. PC1 R^2 peaks at 0.10–0.21 on every model; the linear axis never appears. Helix R^2 peaks at layer 0 on four of the five models (OLMo-3-32B included)—i.e. at the output of the token embedding, before any transformer block has executed (Figure 3). The FFT magnitudes are ~ 0.20 (compare ~ 30 for Latin); the apparent $T = 10$ peak is a tokeniser artifact in which the second Greek letter cycles through $\alpha, \beta, \gamma, \dots, \theta$ every ten numbers. There is no number representation at any layer for the “helix collapse” to collapse *from*.

The two non-positional collapses are not the same failure mode. Roman is encoded as a sum of separately-stored parts; Greek is not encoded quantitatively at all. The standard “helix R^2 ” metric averages over both modes and reports them as comparable; reading PC1 R^2 and its layer-of-peak alongside helix R^2 is what separates them.

4.5 2D PCA: cluster structure made visible (and the PC1-is-tokenization surprise)

To probe non-periodic structure that the FFT and the trig-basis fit cannot see, we added a basis-free diagnostic to every cell: a 2D PCA scatter of $h(a)$ at the helix-peak layer, coloured by a , with each numeral labelled and consecutive integers connected by a faint trajectory line.

Eight scripts $\times$ five models = 40 panels at `out/<model>/<script>/mean/fig4_pca_2d.png`. Three findings fall out.

Roman: cluster-by-prefix is visible directly. The 2D PCA for Pythia/Roman (Figure 4) shows distinct regions of representation space corresponding to the Roman-numeral prefix classes (I/V, X, L, XC), with the trajectory line zigzagging between clusters at the threshold integers ($4 \rightarrow 5$, $9 \rightarrow 10$, $49 \rightarrow 50$, $89 \rightarrow 90$). Roman’s representation is not a manifold but a partitioning of representation space into prefix classes with crude ordering within each class.

Greek: diffuse noise, confirming the no-representation reading. $PC1 + PC2 = 15.5\%$ of total variance on Pythia/Greek (less than half of any other cell) and the points scatter without recognisable shape. There is no learned representation to find; the “helix collapse” was a collapse of nothing.

The Gemma surprise: PC1 is not magnitude; it is token shape. On Gemma-4-31B, $\text{helix}/\text{PCA} = 0.75$ and $\text{helix } R^2 = 0.70$ on Latin—strong by any measure. But PC1 captures **70.9%** of variance (versus Pythia’s 20.4%) and that variance is along token-shape rather than magnitude (Figure 5). Single-digit numerals (0–9) form one cluster, two-digit numerals starting with “1” (10–19) form another, and 20–99 fill a third dense region. The split is by digit count and leading-digit class. The helix lives in lower-variance directions; PC1 alone would not have found it.

The same pattern appears on Devanagari with $PC1 = 94.6\%$ (split between – and –), and on Roman at $PC1 = 96.6\%$ (split at the L threshold). In each case PC1 captures massive variance, but the variance is overwhelmingly tokenization features, not the number line. In Pythia, by contrast, $PC1 R^2$ and *linearity-in-a* correspond cleanly: looking at PC1 is looking at magnitude.

This carries one specific methodological warning: **a high PC1 R^2 is not by itself evidence of a clean magnitude axis**—one must also inspect what PC1 separates. In larger or more multilingual models, tokenization features may dominate the first principal component while the helix occupies smaller-variance subspaces. This also partly explains the depth-variance finding of Section 3: Gemma’s helix shares residual-stream space with strong tokenization signals, so it appears as a sharper spike at a specific layer rather than a broad mid-stack plateau.

5 Babylonian: base 60 is there, once you measure properly

Babylonian cuneiform is positional at base 60, additive within each sexagesimal column. The number 23 is <<!!! (two tens plus three ones); 60 is !* (one in the 60s column, a late-period zero placeholder in the ones column); 599 is !!!!!!!<<<<!!!!!!!. The system has all the ingredients of positional notation, at base 60 instead of base 10.

5.1 The narrow-window measurement says “no”—but cannot

Running the standard $n_{\max} = 100$ protocol with the paper’s basis $T = \{2, 5, 10, 100\}$ gives Table 5.

Table 5: Babylonian on the standard 0–99 window, paper basis.

Model	PC1 R^2	helix R^2	helix/PCA
Pythia-6.9B	0.06	0.71	0.74
Llama-3.1-8B	0.10	0.65	0.69
Gemma-4-E4B	0.40	0.75	0.77
Gemma-4-31B	0.51	0.76	0.77
OLMo-3-32B	0.18	0.73	0.75

A naive read is that Pythia and Llama have circles but no number-line ($\text{PC1 } R^2 \approx 0$), while Gemma has both. Inspecting the FFT (Figure 6) shows labelled peaks at $T = 10$ and $T = 100$, but *no* $T = 60$. The PC1 panel shows two clusters with a hard step at 60—the wrap-boundary is encoded as a categorical state change, not as a rotation.

This is the measurement artifact: in $[0, 99]$, base 60 wraps exactly *once*, at $a = 60$. A single discontinuity is not a periodic signal an FFT can localise. “No base-60 structure” is, at this $n_{\max}$, indistinguishable from “structure the window cannot resolve.”

5.2 Widening the window: $T \approx 60$ emerges in the FFT

At $n_{\max} = 600$ —ten full sexagesimal wraps—the picture changes sharply (Table 6). Every model now has a real magnitude axis ($\text{PC1 } R^2 \approx 0.55\text{--}0.69$); Pythia’s and Llama’s previous $\text{PC1 } R^2 \approx 0$ was an artifact of one wrap.

Table 6: Babylonian, $n_{\max} = 100 \rightarrow 600$ (paper basis $T = \{2, 5, 10, 100\}$). Both PC1 and FFT diagnostics improve; helix R^2 *drops* because $T = 60$ is not in the basis.

Model	PC1 R^2 (100→600)	helix R^2 (100→600)	helix/PCA
Pythia-6.9B	0.06 → 0.55	0.71 → 0.66	0.74 → 0.69
Llama-3.1-8B	0.10 → 0.55	0.65 → 0.52	0.69 → 0.56
Gemma-4-E4B	0.40 → 0.69	0.75 → 0.61	0.77 → 0.64
Gemma-4-31B	0.51 → 0.64	0.76 → 0.50	0.77 → 0.52
OLMo-3-32B	0.18 → 0.62	0.73 → 0.53	0.75 → 0.56

The FFT at $n_{\max} = 600$ shows a clean labelled peak at $T \approx 60$ (Figure 7). For Llama and Gemma-4-31B, base-60 structure appears mostly via its second harmonic at $T = 30$ (frequency $2/60 = 1/30$, Figure 8). A pure sinusoid at $T = 60$ would have only the fundamental; the presence of strong even harmonics indicates a more sawtooth-like underlying signal, consistent with the per-column reset structure of cuneiform.

So the structure is *present in the FFT*—but helix R^2 on the paper basis *drops* (Table 6), because $T \in \{2, 5, 10, 100\}$ has no column against which to regress period-60 variance. The basis is blind to a frequency the data contains.

5.3 An extended basis recovers 7–13 percentage points of variance, on every model

We re-ran the $n_{\max} = 600$ sweep with $T = \{2, 5, 10, 60, 100\}$ —a single additional period, two additional columns ($\cos \frac{2\pi a}{60}, \sin \frac{2\pi a}{60}$). Table 7 reports the result.

Table 7: Babylonian at $n_{\max} = 600$: paper basis $T = \{2, 5, 10, 100\}$ vs. extended basis $T = \{2, 5, 10, 60, 100\}$. Adding the single missing period recovers 7–13 percentage points of variance on every model.

Model	helix R^2 (paper → ext.)	Δ	helix/PCA (paper → ext.)	Δ
Pythia-6.9B	0.661 → 0.747	+0.086	0.687 → 0.772	+0.085
Llama-3.1-8B	0.515 → 0.622	+0.107	0.555 → 0.645	+0.090
Gemma-4-E4B	0.611 → 0.698	+0.087	0.637 → 0.699	+0.062
Gemma-4-31B	0.501 → 0.634	+0.133	0.523 → 0.636	+0.113
OLMo-3-32B	0.532 → 0.619	+0.087	0.559 → 0.645	+0.086

The two models whose FFT showed strongest second-harmonic content (Llama-3.1-8B and Gemma-4-31B) gain the most (+0.107 and +0.133). The regression cannot extract a non-

sinusoidal signal from a basis that lacks its fundamental; providing $T = 60$ anchors the phase and the model’s sawtooth-like representation lands on it. PC1 R^2 is unchanged by the basis change, confirming that no other variable has shifted.

The conclusion is unambiguous:

Base-60 structure is present in the Babylonian representation on all five models, with OLMo-3-32B at Δ helix $R^2 \approx 0.09$ from adding $T = 60$ to the basis—inside the 7–13 pp band the other four models exhibit.

Both measurement adjustments (wider window, extended basis) point the same direction, which is the strongest evidence we have that the structure is real rather than an artifact of either change.

6 Where is the helix built? $L=0$ vs. the peak layer

A helix R^2 value at a model’s peak layer is the headline number, but it conflates two very different things: structure that the model *computes* through its transformer blocks, and structure that the tokeniser + our mean-pooling protocol *already provide* before any block has executed. For every numeral a , the residual stream at $L = 0$ is the mean of the embedding vectors of a ’s sub-tokens. For an additive numeral system—one wedge per unit, or one Roman letter per stroke—this mean is approximately a linear function of the per-symbol counts. The trig basis can then fit it directly, *independently* of what the transformer does on top.

The clean diagnostic is the ratio $\rho_{L0} := \text{helix } R^2(L=0) / \max_L \text{helix } R^2(L)$. $\rho_{L0} \approx 1$ means the transformer adds essentially nothing to the helix fit: depth’s role is at most to preserve, not build, the geometry. ρ_{L0} small means the transformer is doing real representational work. We compute ρ_{L0} for all $5 \times 8 = 40$ cells of our matrix; the result (Table 8) is more revealing than any single per-cell helix R^2 .

Table 8: The ratio $\rho_{L0} = \text{helix } R^2$ at $L = 0$ divided by helix R^2 at the peak layer, for every (model, script) cell. Values near 1 mean the helix is already present in the embedding-layer output; values near 0 mean the transformer blocks build it.

script	Pythia	Llama	E4B	31B	OLMo
latin	0.34	0.39	0.89	0.79	0.63
arabic	1.00	1.00	0.80	0.74	0.92
persian	1.00	0.73	0.63	0.61	0.98
devanagari	0.99	1.00	0.49	0.44	0.85
chinese	0.95	0.99	0.96	0.88	1.00
greek	1.00	1.00	0.56	0.55	1.00
roman	0.78	0.73	0.55	0.48	0.65
babylonian	0.86	0.87	0.89	0.88	0.82

Several patterns matter:

Pythia/Latin and Llama/Latin are the cells where depth genuinely builds the helix. $\rho_{L0} = 0.34$ and 0.39 respectively—i.e. the transformer blocks contribute the majority of the helix fit, with the embedding layer providing less than half. These are the exact cells the original paper [1] studies, and our ρ_{L0} analysis confirms that on those cells the helix is a genuinely computed representation, not a tokeniser artifact.

The cross-script extension on Pythia and Llama is largely a tokeniser artifact. Every non-Latin positional script on Pythia and Llama has $\rho_{L0} \in [0.73, 1.00]$. Arabic-Indic and Greek on both models hit 1.00 exactly: the helix- R^2 peak *is* the embedding layer. The transformer never adds anything. What our Finding in Section 4 initially called “the helix transfers to other positional scripts on multilingual models” becomes, more honestly: *the tokeniser preserves enough digit-level structure that mean-pooling produces a helix-fittable representation, regardless of whether the model uses it for anything*. Pythia is not multilingual; its Arabic “helix” R^2 of 0.44 at $L=0$ comes from the tokeniser splitting Arabic-Indic numerals into digit-level sub-tokens that the trig basis can then fit.

Gemma builds the helix where it has been trained to. Gemma-4-E4B and Gemma-4-31B have $\rho_{L0} \in [0.44, 0.63]$ on Devanagari, Persian, and Roman. Those are the cells where the transformer is doing the most representational work in our matrix. This is consistent with Gemma 4’s heavy Indic-language and classical-language training mix: depth processes those scripts into a genuine quantitative encoding, while less-represented scripts (Chinese, Greek) remain tokeniser-driven. Gemma’s high “PC1 = tokenisation” figures from Section 4.5 are the same phenomenon read along a different axis.

OLMo sits between Pythia/Llama and Gemma. OLMo’s Latin $\rho_{L0} = 0.63$ is intermediate—some helix at $L = 0$, but the transformer still does roughly a third of the work. Its *multilingual* cells (Arabic 0.92, Persian 0.98, Devanagari 0.85) are closer to the Pythia/Llama pattern: tokeniser-driven. OLMo exhibits the Gemma-style early tokenisation structure on Latin while treating non-Latin scripts more like Pythia and Llama treat them.

Babylonian is universally tokeniser-driven. $\rho_{L0} = 0.82\text{--}0.89$ across every model. The transformer contributes at most ~ 15 pp on top of what the wedge-token embeddings + mean-pooling already supply. The strong Δ helix R^2 from adding $T = 60$ to the basis (Section 5.3) is then most honestly read as *the trig basis fitting the tokeniser-pooled representation of cuneiform glyph counts*, not as the model having learned base-60 arithmetic from its training corpus. The result is still nontrivial—the wedge-token embeddings must already carry value information for the pooling to produce this structure—but the cause is the embedding layer, not the transformer.

Revised reading of the cross-script and Babylonian findings: The cleanest evidence for “the model represents numerals as a helix” is the Pythia/Latin and Llama/Latin cells of the original paper, plus the Gemma cells on Devanagari, Persian, and Roman, and the Gemma/Latin cell. Most other cells in our matrix—including all Babylonian cells—report a helix R^2 that is largely the tokeniser plus our pooling protocol.

7 Method matters: two general lessons

M1. The basis has bandwidth. Helix R^2 is a fitted regression onto a fixed set of periods. It is sensitive only to structure at those periods; if a model encodes something at a period the basis omits, helix R^2 will be low even when the underlying structure is strong. The Babylonian result shows this concretely: same data, same model, $+0.13$ helix R^2 from adding one period (Table 7). The fix is to *always cross-check a basis-fit metric against a basis-free one*. If the FFT shows a peak at a period not in the basis, helix R^2 is undermeasuring the structure; add the period and refit.

M2. The window must wrap. To detect a period T via FFT, the input range must contain several wraps of T . One wrap is a single discontinuity, not a periodic signal. The 0–99 window is fine for $T \in \{2, 5, 10\}$ (50, 20, 10 wraps) and barely workable for $T = 100$ (one wrap, visible as a slow trend). It is *not* workable for $T = 60$: one wrap, in a phase-misaligned position relative to the window’s endpoints, produces noise rather than a trend. The rule is to set $n_{\max}$ to a large multiple of every candidate period.

These two corrections compose: on Babylonian at $n_{\max} = 100$ with the paper basis, both blind spots conspire in the same direction (“no base-60 here”). Fixing only one and not the other gives partial evidence; fixing both, and observing that the metrics move together and in the predicted direction, is what makes the inference robust.

8 Discussion and conclusion

Putting the threads together:

- **Cross-architecture:** the helix appears on Gemma-4-E4B and Gemma-4-31B—a new model family beyond the three (GPT-J, Pythia, Llama) the original paper tested—and re-measurement on Pythia and Llama under our wider protocol replicates the helix at very different depths and with qualitatively different layer-profiles. “Read the middle layer” is a Pythia/GPT-J convention. Find-the-layer is a necessary step for every new model.
- **Cross-script, positional:** the geometric encoding transfers across glyph systems for any positional base-10 script the model has been trained on. Multilingual models reproduce the helix on Devanagari and Arabic-Indic; English-heavy models do not. The encoding is built per-script in pre-training; positional structure is necessary but not sufficient.
- **Cross-script, non-positional:** Greek and Roman both collapse the helix metric, but for different reasons. Roman splits into a number-line module and a letter-cycling module living in different layers; Greek never forms a quantitative representation at all (the only periodic structure in the FFT is a tokeniser artifact at $L = 0$). The two failure modes are distinguishable only by reading PC1 R^2 and its layer-of-peak alongside helix R^2 . The 2-D PCA scatter (Section 4.5) further reveals that Roman is encoded as discrete prefix-class clusters with crude ordering within each, and surfaces the most surprising finding of the script axis: on Gemma the first principal component of the residual stream captures up to 96.6% of variance but represents *tokenisation* (digit-count, leading-digit class), not magnitude. High PC1 R^2 is not by itself evidence of a magnitude axis; the helix in Gemma lives in lower-variance directions.
- **Cross-script, sexagesimal:** Babylonian cuneiform’s base 60 is fittable by an extended trig basis on all five models, with Δ helix R^2 from adding $T = 60$ in the +0.087 to +0.133 range. Recovering it required widening the window past one wrap and adding $T = 60$ to the trig basis.
- **Where the helix is built** (Section 6): the L0-vs-peak diagnostic reveals that most of our cross-script and Babylonian helices are already present at the embedding layer, before any transformer block has run ($\rho_{L0} = 0.82$ –1.00 on every Babylonian cell and on every Pythia/Llama non-Latin positional cell). The cells where depth genuinely builds the helix are Pythia/Latin and Llama/Latin ($\rho_{L0} = 0.34, 0.39$) and the Gemma cells on Devanagari, Persian, and Roman ($\rho_{L0} = 0.44$ –0.65). The cross-script and Babylonian generalisations are largely artifacts of tokenisation + our mean-pooling protocol, not learned arithmetic representations.

We started this work expecting the helix to be a property of one model’s training. What we found is sharper and less flattering: the helix is a property of how transformers *can* encode

quantity when there is one to encode, but on most cells in our matrix the “helix” we measure is already in the tokeniser-pooled embeddings before any block has run. The original paper’s Pythia/Latin cell is where depth genuinely builds the geometry; the cross-script extension on Pythia and Llama is largely the tokeniser; Gemma’s Indic and classical scripts are the cells outside Latin where depth most clearly takes over. Whether the helix subspace is also *the representation that downstream computation reads* is a causal question this paper does not answer; the natural intervention sits at the answer position rather than the operand position, and we leave it to follow-up work.

8.1 Follow-ups

The natural next experiments are causal:

1. **Helix-subspace patching at the answer position.** The original paper’s strongest causal claim is that on Pythia/Latin, patching only the helix-basis subspace (~ 9 directions of several thousand) at the position whose hidden state produces the answer is sufficient to control the answer. Replicating that on Pythia and extending to Llama, Gemma-4-E4B and Gemma-4-31B—and to the non-Latin positional scripts where the geometric helix is present in this paper—is the most direct causal follow-up to the work reported here.
2. **Wider basis.** The helix basis used here is the paper’s $T \in \{2, 5, 10, 100\}$. Extending to the basis that recovers the Babylonian base-60 finding ($T \in \{2, 5, 10, 60, 100\}$), or to a *learned* subspace whose rank is set by cumulative explained-variance rather than by a prescribed period list, is the natural follow-up if a four-period helix-subspace intervention leaves a large unexplained fraction of the causal effect.
3. **Sexagesimal rotation.** On Gemma-4-31B’s Babylonian representation at $n_{\max} = 600$, rotating only the $(\cos 2\pi a/60, \sin 2\pi a/60)$ direction pair should shift the answer-token logit distribution by exactly 60 if the model uses the base-60 axis computationally. This is the cleanest test of representation-vs-use for a non-base-10 period.
4. **Cross-layer units-digit rescue.** For the E4B/Latin dissociation: project residual at L_5 onto $(\cos 2\pi a/10, \sin 2\pi a/10)$ and inject the projection at a later layer to see whether the units-digit information is discarded by downstream layers or remains accessible.

Secondary follow-ups suggested by the descriptive work but not addressed here: Greek alphabetic numerals plausibly form 9 clusters of 10, one per tens-letter, visible only via 2-D PCA scatter (the diagnostic is included but cluster analysis is left for future work); Roman’s encoding looks like a piecewise-linear staircase with discontinuities at glyph thresholds, which an additive regression $h(a) \approx \sum_{\text{symbol}} \text{count}(\text{symbol}) \cdot u_{\text{symbol}}$ would test directly; and the layer at which the Babylonian base-60 representation peaks, and how it differs from the Latin base-10 peak in the same model, is unaddressed in this paper and worth dedicated analysis.

Reproducibility

All code, configuration, and figures are in the public repository accompanying this paper. Each experiment lives in its own driver script with PEP 723 inline metadata, runnable via `uv run`; each has an accompanying `./run_*.sh` bash wrapper that loops over the five models and is idempotent (each combination is skipped if its expected output file already exists).

Representational sweep (Sections 3–5). `./run.sh` drives 50 combinations across 5 models, 8 scripts, 2 windows ($n_{\max} \in \{100, 600\}$), 2 bases ($T = \{2, 5, 10, 100\}$ and $T = \{2, 5, 10, 60, 100\}$):

```

# Default: Pythia-6.9B, Latin digits, paper basis, n=100
uv run main.py

# Cross-script
uv run main.py --model meta-llama/Llama-3.1-8B --script devanagari
uv run main.py --model google/gemma-4-31B --script babylonian --sweep

# Wider window
uv run main.py --script babylonian --n_max 600 --sweep

# Extended basis with T = 60
uv run main.py --script babylonian --n_max 600 \
    --periods 2,5,10,60,100 --sweep

```

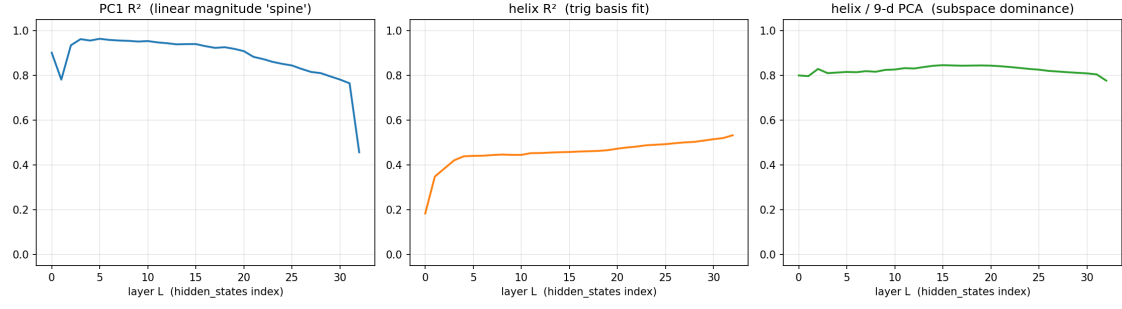
Hardware: Apple M5 Max, 128 GB unified memory (MPS); falls back to CUDA, then CPU.
Total wall-time for the representational sweep: ~ 8 hours.

References

- [1] S. Kantamneni and M. Tegmark. Language Models Use Trigonometry to Do Addition. *arXiv preprint* arXiv:2502.00873, 2025.
- [2] N. Nanda, L. Chan, T. Lieberum, J. Smith, and J. Steinhardt. Progress Measures for Grokking via Mechanistic Interpretability. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2301.05217.
- [3] A. Power, Y. Burda, H. Edwards, I. Babuschkin, and V. Misra. Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets. *arXiv preprint* arXiv:2201.02177, 2022.

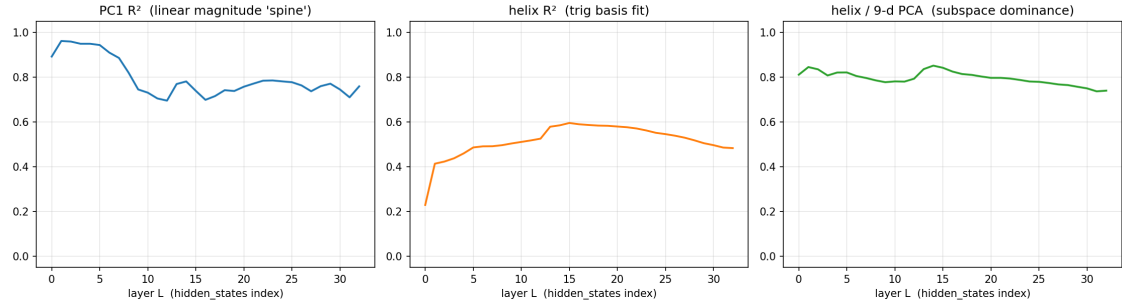
Pythia-6.9B (32 layers)

Layer sweep — EleutherAI/pythia-6.9b / latin / pool=mean / periods=[2, 5, 10, 100]



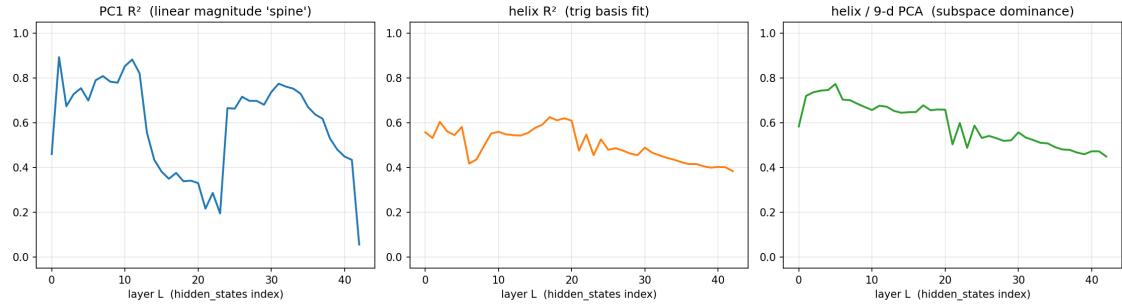
Llama-3.1-8B (32 layers)

Layer sweep — meta-llama/Llama-3.1-8B / latin / pool=mean / periods=[2, 5, 10, 100]



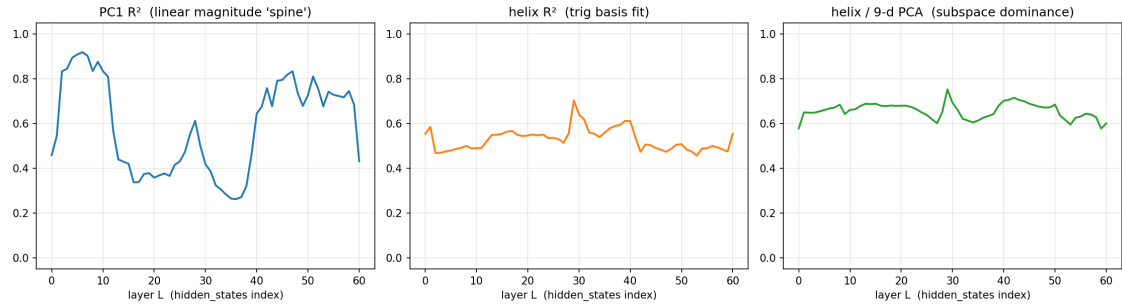
Gemma-4-E4B (34 layers)

Layer sweep — google/gemma-4-E4B / latin / pool=mean / periods=[2, 5, 10, 100]



Gemma-4-31B (61 layers)

Layer sweep — google/gemma-4-31B / latin / pool=mean / periods=[2, 5, 10, 100]



OLMo-3-32B (64 layers)

Layer sweep — allenai/OLMo-3-1125-32B / latin / pool=mean / periods=[2, 5, 10, 100]

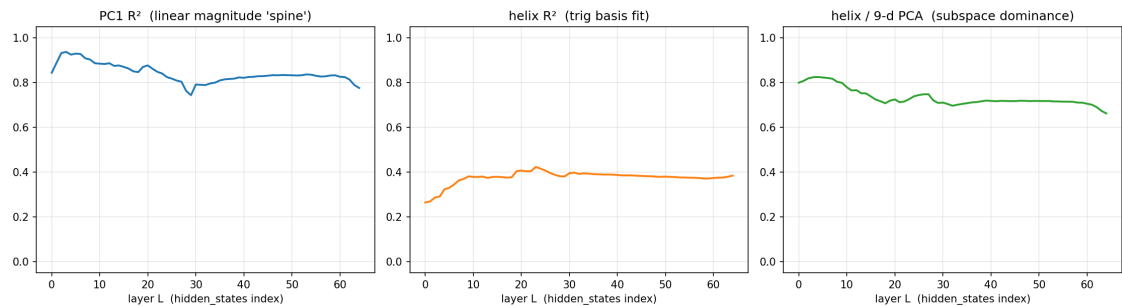


Figure 1: PC1 R^2 , helix R^2 and helix/PCA ratio plotted against layer index, for each of the five models (Latin digits 0–99). Pythia and Llama rise into the back half of the stack and stay high. Gemma-4-E4B peaks early and decays. Gemma-4-31B is nearly flat at ~ 0.5 with a sharp dip at layer 20. OLMo-3-32B is nearly flat at ~ 0.8 with a sharp dip at layer 20.

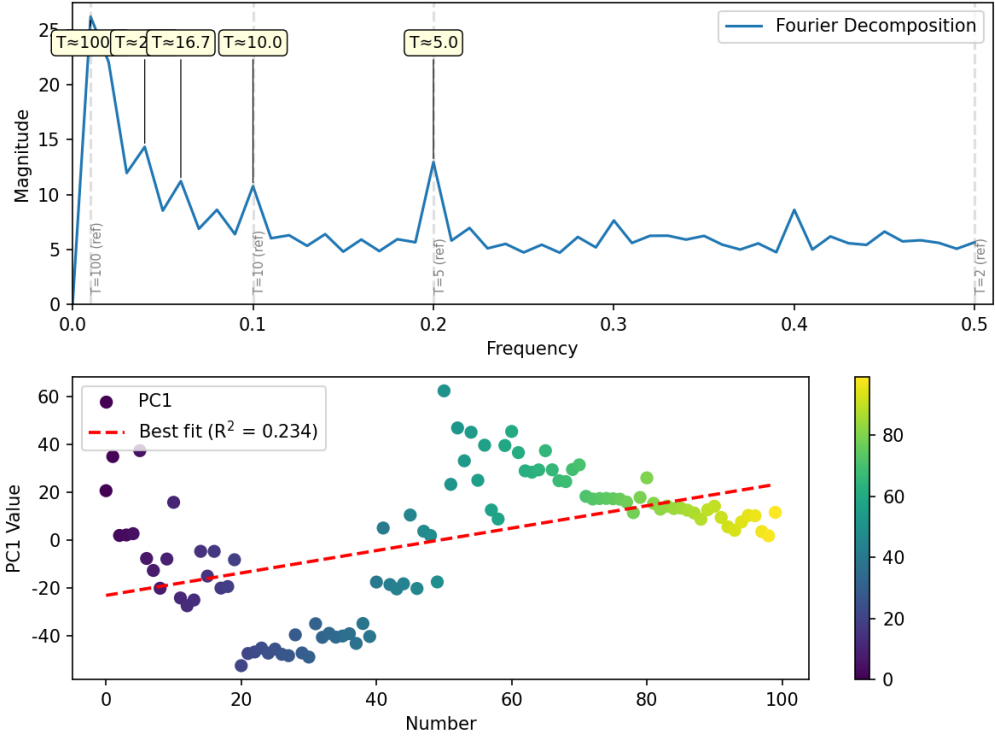


Figure 2: Pythia-6.9B on Roman numerals. **Top:** FFT spectrum shows clean peaks at $T = 10$ (X cycle) and $T = 5$ (V cycle)—the visual periodicity of the additive system. **Bottom:** PC1 versus a is a piecewise function with jumps at the Roman threshold values, not a line. The two components live in different layers (Table 4).

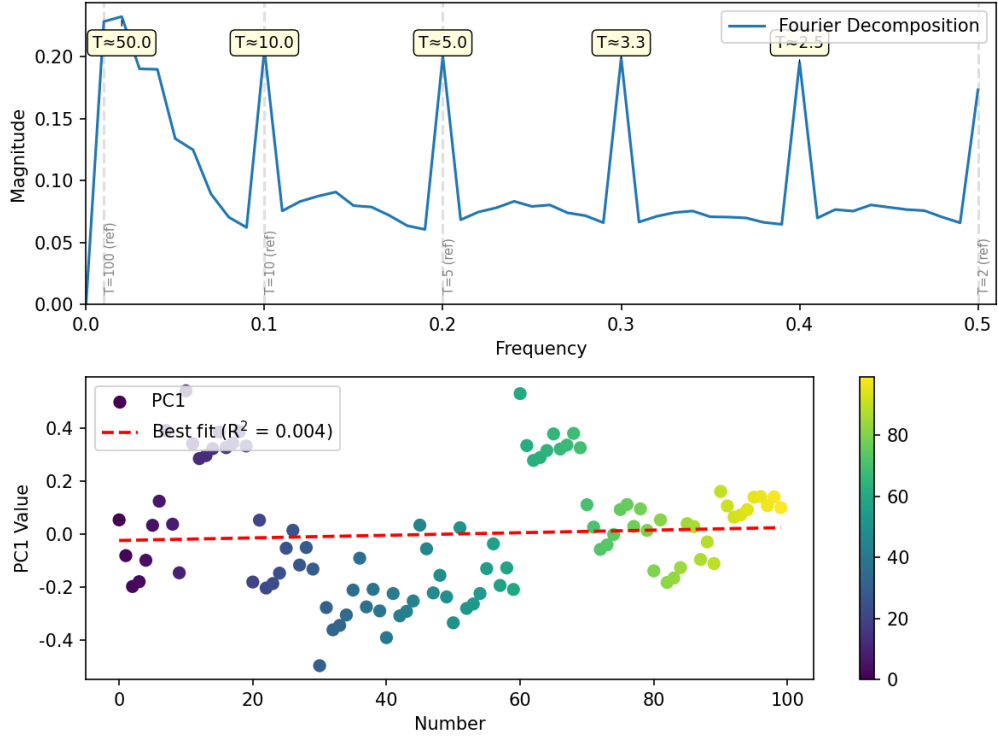


Figure 3: Pythia-6.9B on Greek alphabetic numerals. The FFT magnitude is two orders of magnitude smaller than on Latin; the only periodic structure is a $T = 10$ artifact at $L = 0$ from the alphabetic construction $\kappa + \gamma = \kappa\gamma$. PC1 $R^2 = 0.004$: the model is not representing Greek numerals as quantities.

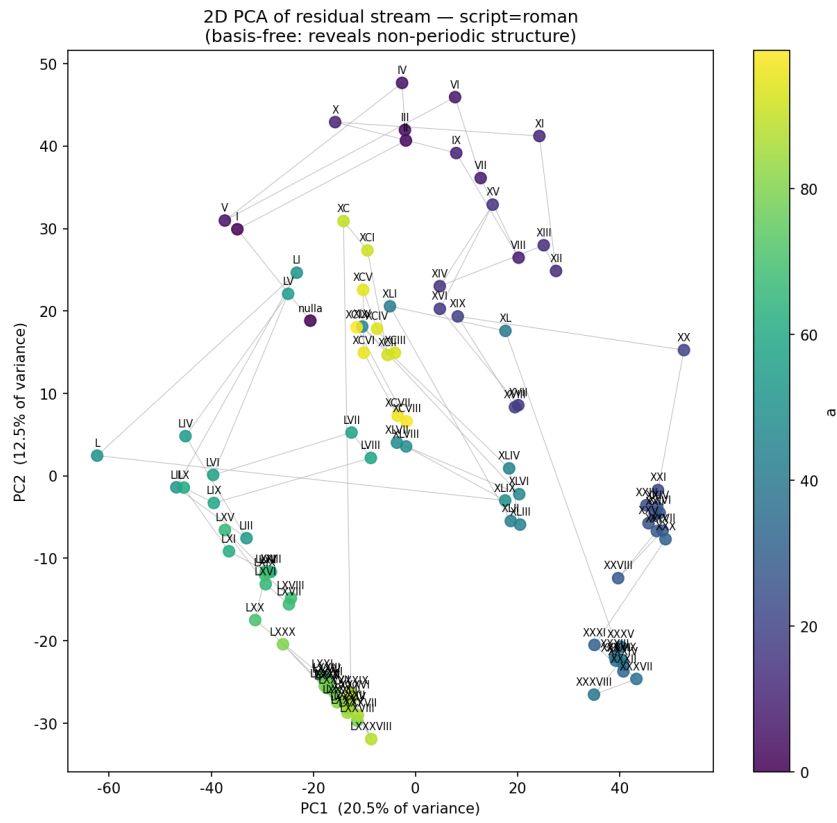


Figure 4: Pythia-6.9B on Roman numerals. 2D PCA at the helix-peak layer reveals discrete clusters by Roman-numeral prefix (I/V, X, L, XC), with the trajectory line zigzagging at threshold integers. The representation is a partition, not a continuous manifold.

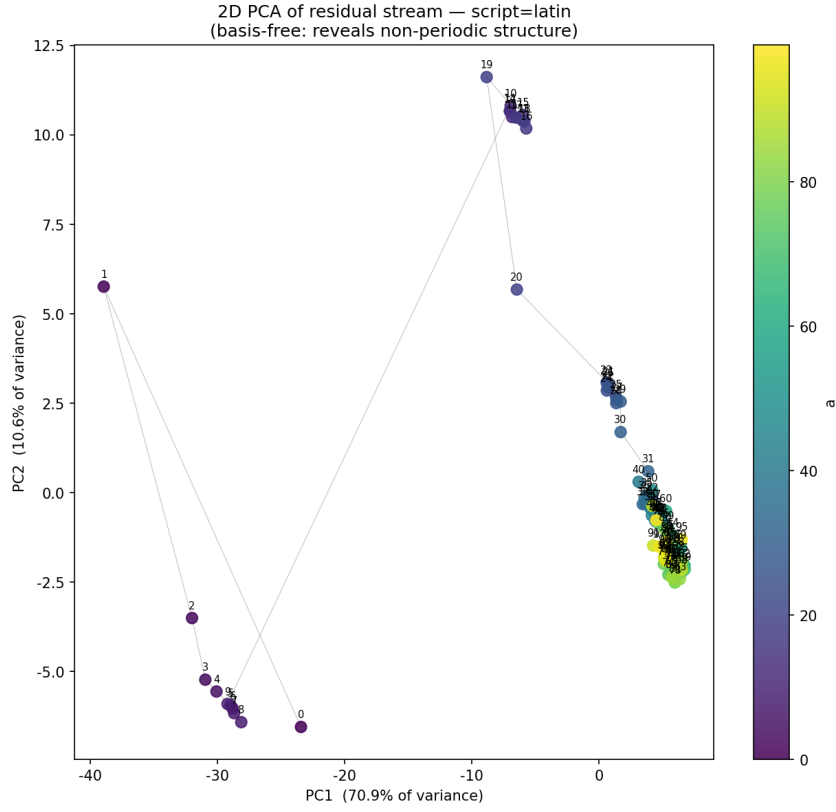


Figure 5: Gemma-4-31B on Latin digits. PC1 captures 70.9% of variance but separates the points by *token shape* (single-digit vs. teens vs. 20–99), not magnitude. Helix $R^2 = 0.70$ persists at this layer, in lower-variance directions PC1 does not reach.

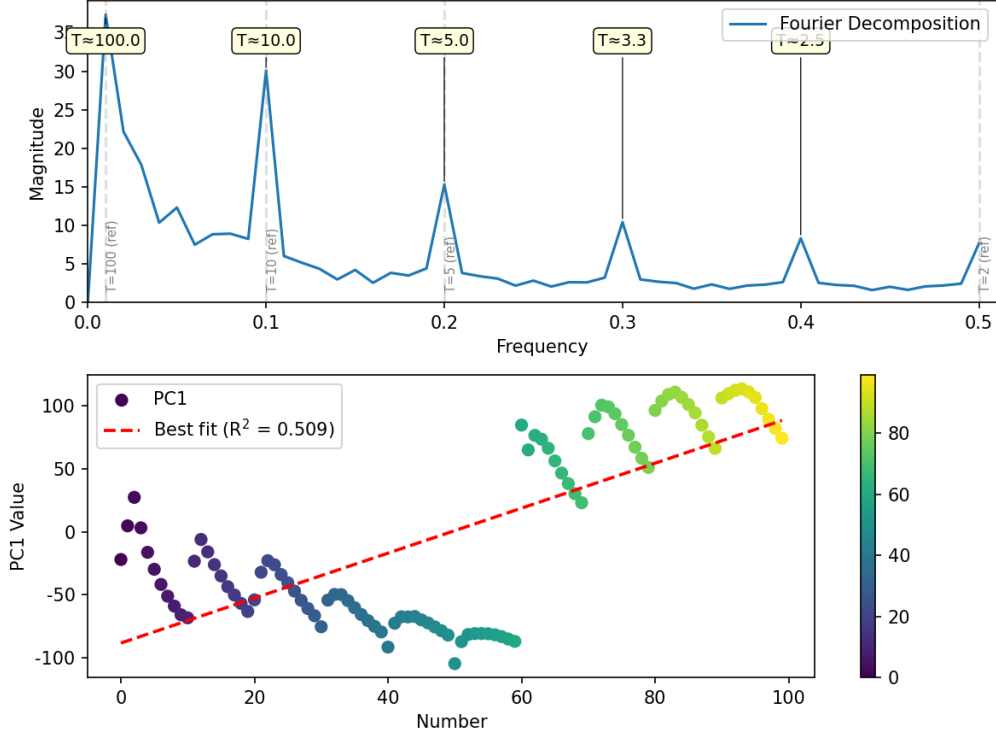


Figure 6: Gemma-4-31B on Babylonian at $n_{\max} = 100$. Top FFT shows $T = 10$ and $T = 100$; *no* $T = 60$. PC1 (bottom) shows numbers 0–59 versus 60–99 as two clusters with a step between them.

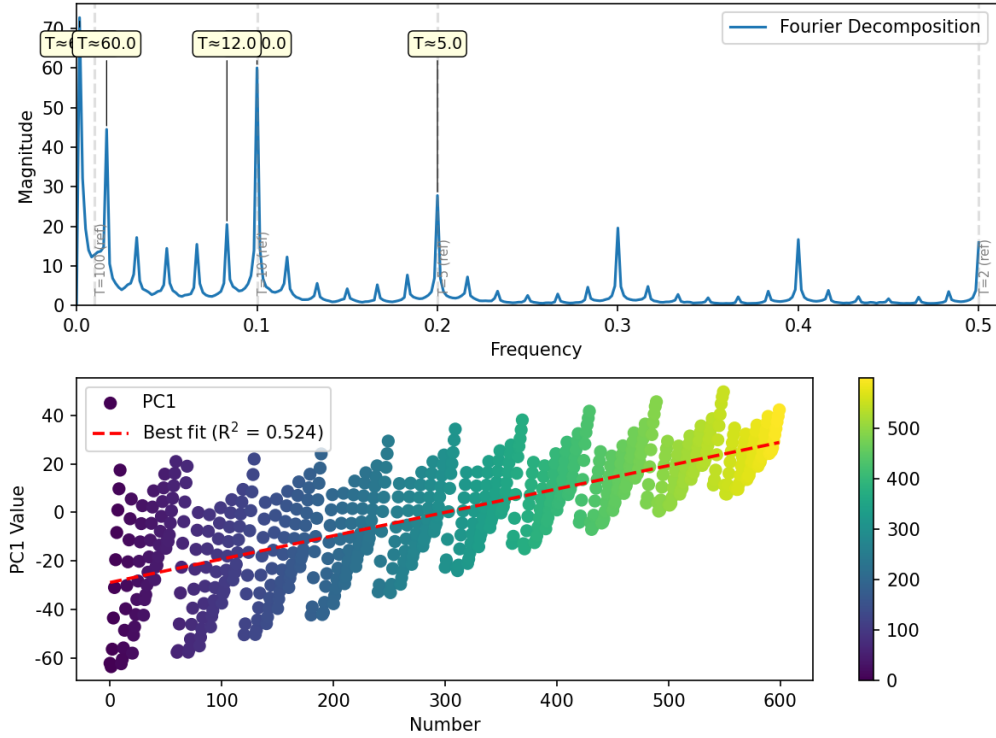


Figure 7: Pythia-6.9B on Babylonian at $n_{\max} = 600$. A clean labelled peak at $T \approx 60$ appears alongside $T = 10$ and various harmonics. The PC1 panel shows the magnitude axis with a per-column sawtooth—the model resets into a new sexagesimal column every 60 integers.

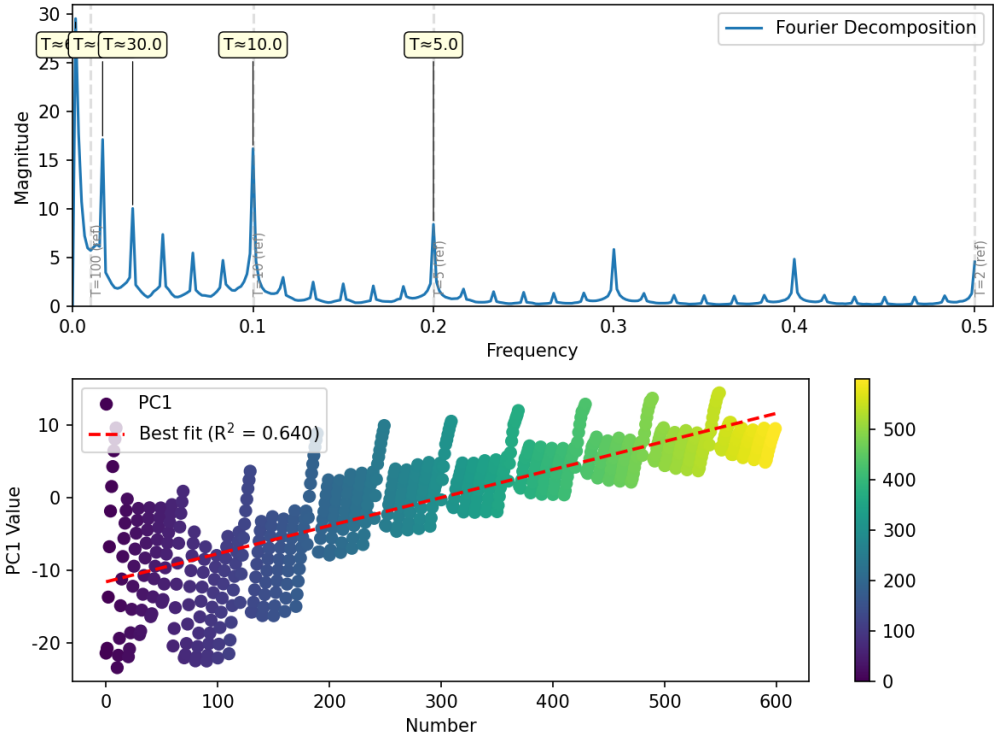


Figure 8: Gemma-4-31B on Babylonian at $n_{\max} = 600$. Base-60 structure manifests primarily through its second harmonic at $T = 30$, indicating a sawtooth-like signal rather than a pure sinusoid.